THE LAB — DAY 09

Machines That Act

Reinforcement learning — learning from reward, not answers

Day 09 · Week 2 **Unlocks 🎮 Game Master**

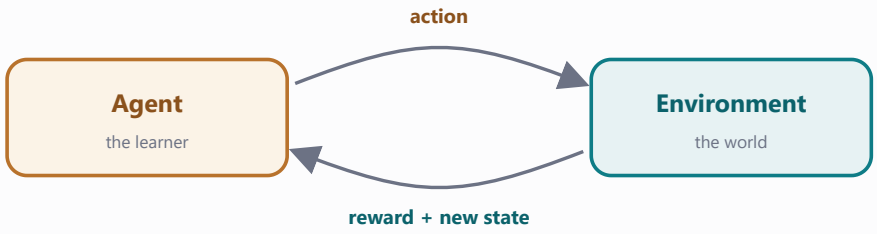
1 Mission briefing

Every model so far learned from examples a human labeled. Today's machine learns with *no* examples at all — only rewards. It tries, fails, and updates its own instincts. You'll teach an agent to cross a frozen lake with a 64-number Q-table, then scale up to a neural network that balances a pole it has never seen. *Previously:* you unlocked 💬 Language Lab — tokens, attention, and next-token prediction.

- ☐ Trace the **RL loop**: agent, action, environment, reward
- ☐ Balance exploring vs exploiting with **ε-greedy**
- ☐ See when tables explode → deep Q-networks (**DQN**)
- ☐ Build a **Q-table** — a 64-number instinct
- ☐ Apply the one-line **Q-learning law**
- ☐ Compare before / after CartPole videos

2 Field guide the concepts

The RL loop. An **agent** in an **environment** picks an **action**, receives a **reward** and a new **state**, and repeats. There is no answer key — only the score it is trying to grow.



The loop: the agent picks an action; the environment answers with a reward and a new state. Repeat thousands of times.

The Q-table is a tiny instinct. For FrozenLake there are 16 tiles × 4 moves = 64 numbers, each saying "how good is this move from here?" **ε-greedy** mostly takes the best-known move (exploit) but a fraction ε of the time tries something random (explore). And the whole thing learns from one line — the **Q-learning law**: `new = old + α(reward + γ·best_future − old)`, nudging each estimate a step toward reality.

When tables explode → DQN. CartPole's state is continuous — infinite rows. So replace the table with a neural network that *predicts* Q-values. Same idea, far bigger brain.

> 🧮 **Matrix Watch —** your Q-table is a 16×4 matrix, and the DQN that replaces it is just layers of matrix multiplies predicting the same Q-values. Same operation, bigger — and high-performance computing makes it fast.

3 Lab missions check each off in the notebook

1 How far does pure luck get? ☐

Run a purely random agent on FrozenLake many times.

Expect: it reach the gift only rarely — luck is a bad strategy.

2 Explore vs exploit (choose_action) ☐

Pick a random move with probability ε, else `argmax(q[s])`.

Expect: the agent wander early and act decisively later.

3 Start bold, end wise (get_epsilon) ☐

Decay ε from near 1 toward near 0 over training.

Expect: exploration high at first, then fading as it learns.

4 Train the agent (the crown jewel) 👑 ☐

Apply the one-line Q-update across thousands of episodes.

Expect: the Q-table sharpen until it solves the lake.

5 How good did it get? ☐

Evaluate the greedy policy's success rate.

Expect: near-100% on the non-slippery lake.

6 Read out the policy ☐

Print the best action per state as arrows.

Expect: a little arrow map that walks straight to the goal.

7 The CartPole baseline ☐

Compare an untrained agent to your trained DQN.

Expect: a few wobbly steps vs a pole balanced toward 500.

4 Cheat sheet Q-learning & Gym

`gym.make("FrozenLake-v1", is_slippery=False)`	Create the frozen-lake world.
`s, info = env.reset(seed=0)`	Start a fresh episode.
`s, r, term, trunc, info = env.step(a)`	Take action a; get the 5-tuple back.
`np.argmax(q[s])`	Greedy move: the best-known action from state s.
`q[s,a] += α*(r + γ*q[s2].max() - q[s,a])`	The Q-learning law, in one line.
`DQN("MlpPolicy", env).learn(60_000)`	A neural network replaces the table.
`model.save("game_master_dqn.zip")`	Freeze the trained agent to a file.
`imageio.mimsave("run.gif", frames, fps=25)`	Turn rendered frames into a replay GIF.

5 🔓 Unlock your Studio module

🎮 Game Master

Save your Q-table and your trained DQN. The unlock cell must write exactly:

`ai_studio/models/game_master_qtable.npy` `ai_studio/models/game_master_dqn.zip`

Watch both agents replay live — the icy walk and the balancing pole:

```
$ python ai_studio/app.py
```

6 Mini-glossary

agent
The learner that chooses actions.

environment
The world that answers with rewards and states.

state
A snapshot of the situation the agent is in.

action
A move the agent can make.

reward
The score signal the agent tries to maximize.

policy
The agent's strategy for choosing actions.

Q-value
The estimated long-term value of an action in a state.

ε-greedy
Mostly exploit the best move, sometimes explore a random one.

DQN
A deep Q-network — a neural net replacing the Q-table.

7 Show & tell

1. Walk us through your Q-learning update line — what does each term actually do?
2. What ε did you use, and what happened when you set ε = 0 (pure greedy)?
3. Show random vs trained CartPole. How many steps did each survive, and why the gap?